Points to Remember

- When current passes through a wire a magnetic field is set up around the wire. This is called magnetic effect of current.
- The space surrounding a bar magnet in which its influence in the form of magnetic force can be detected, is called magnetic field.
- The path along which a free magnetic north pole will move in a magnetic field is called magnetic field lines.
- The magnetic field set up by a current carrying conductor is always at right angles to the direction of flow of current.
- Two parallel wires carrying current in the same directions attract each other.
- Two parallel wires carrying current in the opposite directions repel each other.
- Direction of the force in a current carrying conductor is determined by Fleming's Left Hand Rule.
- Electric motor is a device which converts electrical energy into mechanical energy.
- The phenomenon of producing induced current in a closed circuit due to the change in magnetic field in the circuit is known as electromagnetic induction.
- Direction of induced current in a conductor is determined by Fleming's Right Hand Rule.
- Electric generator is a device used to convert mechanical energy into electrical energy.
- Electric generator works on the principle of electromagnetic induction.
- Transformer converts low alternating current to high alternating current and vice versa.

GLOSSARY

Magnetic field	The region surrounding a magnet in which the force of the magnet can be detected.
Magnetic line of force	The path followed by a magnetic needle in a magnetic field.
Dynamo	Device which converts mechanical energy into electrical energy.
Motor	Device which converts electrical energy into mechanical energy.
Electromagnetic induction	The phenomenon of producing an induced emf due to change in the magnetic lines of forces associated with a conductor.
Transformer	Device which converts low alternating current to high alternating current and vice versa.
MRI	Devise used to obtain images of the internal parts of our body.

TEXTBOOK EXERCISES

YGPJ2

I. Choose the correct answer.

1. Which of the following converts electrical energy into mechanical energy?

   a) Motor b) Battery
   c) Generator d) Switch

2. Transformer works on

   a) AC only b) DC only
   c) Both AC and DC

3. The part of the AC generator that passes the current from the armature coil to the external circuit is

   a) field magnet b) split rings
   c) slip rings d) brushes

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